

First-round interview — recruitment agency screen

The questions below are only recommendations, you can use your own set of questions that cover related topics. You do not need to ask them all, few questions in topic should be enough to understand the candidate's knowledge. Sections that can't be skipped: Java memory model, classes and interfaces, collections, concurrency, Spring, JPA, databases.

Estimate each section with a ranking:

5 => Absolute Star

4 => Excellent overall

3 => Meets requirement

2 => Doesn't meet requirement, but has few bright spots

1 => Not good enough

Java memory model

- What kinds of memory exist in Java? What is the difference between them? Which is used when?
- What is the stack for, and what is stored in it?
- What regions is the heap divided into?
- Explain garbage collection in Java.
- Which garbage collectors do you know?
- How does the collector know an object has become garbage?
- Do primitives need to be collected?
- How does G1 work?

Data types

- The two main groups of data types
- How to compare primitives and objects
- Passing method arguments by value vs by reference

Object o = new Object();

- What is stored in o?
- Where does new Object() live?

Boxing/unboxing

Strings in Java

- Is there a difference between String s1 = "string" and String s2 = new String("string")?
- s1 == s2 vs s1.equals(s2)
- String.intern()
- Are String objects mutable or immutable?

Inheritance and generics

- What are generics, and what are they for?
- Examples of generics in Java
- extends / super (in generic bounds and wildcards)

Abstract classes and interfaces

- Difference between an abstract class and an interface
- How many supertypes you can extend / implement

- Presence of a constructor
- Default access modifiers of abstract class and interface
- Multiple inheritance with interfaces
- What if two interface methods “overlap” (same signature from two interfaces)?
- What if two default methods in interfaces conflict?

Collections

- What kinds of collections exist in Java?
- Describe List.
- Time complexity of insertion and read/access.
- When is an array-based list better, when is a linked list better?
- How does a list differ from a plain array?
- What is an iterator?

HashMap

- Describe HashMap.
- How is it implemented?
- What is a hash code?
- Hash code normalization (or: spreading / masking the hash).
- How is it related to equals?
- Why does that matter for a hash map?
- Data structure in a bucket when there is a collision.
- What are capacity and load factor? What is the default load factor?
- Time complexity of lookup and insertion.
- In what order do you get elements if you iterate the map? Why?
- Difference between HashMap and TreeMap.

Streams

- In Java, what is a Stream?
- What is the difference between intermediate and terminal stream operations? Give two examples of each.
- After you call a terminal operation on a Stream, can you run another terminal operation on the same Stream reference?
- Does Optional have terminal/intermediate operations like a stream, or is it a one-shot container for zero or one value?
- Besides Stream<T>, which primitive-specialized stream types exist?
- Difference between map and flatMap on a stream
- What is a lambda expression in Java? What is it compiled to (functional interface, syntactic sugar)?
- Standard functional interfaces

Concurrency

- How do you safely share mutable state (field, collection, counter) across threads?
- What goes wrong if two threads read–modify–write a shared variable without coordination?
- What is a data race? Is the outcome always obviously wrong?
- Why doesn’t making a field volatile alone make i++ atomic?
- How does CAS in AtomicInteger differ from i++ under synchronized?

- When prefer atomics vs locks / synchronized?

ReentrantLock

- What does reentrant mean here?
- Why must unlock run in finally?
- What do synchronized and ReentrantLock share regarding mutual exclusion?
- What can ReentrantLock do that synchronized cannot (e.g. interruptible wait, tryLock, timed lock, multiple conditions)?
- Can the same thread re-enter synchronized on the same monitor — why?

Spring

- What is a Spring bean?
- Where are beans stored, and how do you retrieve them from the container?
- What ways exist to inject dependencies (constructor, setter, field, etc.)?
- Which bean scopes do you know (singleton, prototype, request, session, ...)?
- What is @PostConstruct used for?
- What is @Qualifier used for?
- What is @Primary used for?
- What is @Order used for (e.g. ordered beans, @Configuration classes, filters, aspects)?
- Can Spring run code after the web application has finished starting up? (If yes, how?)
- What is a circular dependency between beans, and how can you resolve or avoid it?
- Which HTTP methods do you know?
- How do you understand cookies, HTTP headers, and HTTP session?
- What is CORS (Cross-Origin Resource Sharing)?
- What is idempotency (in the context of HTTP APIs)?
- Spring data JPA/Hibernate
- Hibernate fetch types (EAGER, LAZY)
- @OneToOne, @OneToMany, @ManyToMany
- N+1 problem

Databases

- What is optimistic locking? How is it implemented in databases?
- What is pessimistic locking? How is it implemented?
- Which of them would you prefer and in what cases?
- What is a DB index.
 - What is the purpose of the index?
 - Pros and cons of having the index? If no cons - why not create index for all possible combinations of the fields
 - Explain briefly how btree index works

Design patterns

- Name the three main groups of design patterns.
- Name one pattern from each group.
- Which three patterns do you most often see in Spring?